PRATIKUM JARINGAN KOMPUTER



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I. PENDAHULUAN

1.1 Tujuan Pembelajaran

1. Mahasiswa mampu memahami proses transmisi data menggunakan model TCP/IP.
2. Mahasiswa mampu mengidentifikasi layer-layer pada model TCP/IP.
3. Mahasiswa mampu menggunakan Wireshark untuk menangkap dan menganalisis data jaringan.
4. Mahasiswa mampu mengidentifikasi jenis protokol jaringan berdasarkan data yang ditangkap pada setiap layer TCP/IP.

1.2 Dasar Teori

TCP/IP (Transmission Control Protocol/Internet Protocol) merupakan kumpulan protokol yang digunakan untuk mengatur proses komunikasi dan pertukaran data antar komputer dalam jaringan. TCP/IP menggunakan arsitektur berlapis yang terdiri dari empat layer, yaitu Application Layer, Host-to-Host Layer, Internetwork Layer, dan Network Interface Layer. Modul 2. TCP-IP Model (1)

Setiap layer pada model TCP/IP memiliki fungsi yang berbeda dalam proses komunikasi data. Model TCP/IP juga dapat dipetakan dengan model OSI yang memiliki tujuh layer. Pembagian layer tersebut membantu proses komunikasi data menjadi lebih terstruktur, sehingga setiap bagian dapat menjalankan fungsi tertentu dalam proses pengiriman dan penerimaan data. Modul 2. TCP-IP Model (1)

Dalam praktikum ini digunakan Wireshark sebagai aplikasi untuk menangkap dan menganalisis paket data yang melewati jaringan. Wireshark dapat menampilkan informasi seperti alamat IP pengirim dan tujuan, jenis protokol, panjang data, serta informasi paket yang ditangkap. Data tersebut kemudian dapat difilter berdasarkan protokol tertentu seperti ICMP, DNS, FTP, dan HTTP untuk melihat proses komunikasi pada jaringan.

1.3 Alat dan Bahan

1. Modul praktikum TCP/IP Model.
2. laptop yang telah terinstal aplikasi Wireshark.
3. Jaringan aktif.
4. Web browser.
5. Lembar laporan praktikum.

II. TUGAS

2.1 Melakukan rekam data untuk jenis protokol tertentu

1. Lakukan praktikum seperti kegiatan 2.5.1 dari nomer 4 s/d 8.
2. Ganti kegiatan pada no 7, dengan kegiatan berikut:
    a. Jalankan aplikasi terminal dan ketik perintah NSLOOKUP yang digunakan untuk mengetahui IP dari suatu tujuan (tanpa tanda $):

        $ nslookup www.pens.ac.id

        Ambil data untuk jenis tipe: dns

    b. Jalankan aplikasi web browser dan arahkan ke alamat:

        ftp://newfs.pens.ac.id

        Download salah satu data yang ada di alamat tersebut !!!

        Ambil data untuk jenis tipe: ftp

    c. Jalankan aplikasi web browser dan arahkan ke alamat:

        http://www.pens.ac.id

        Ambil data untuk jenis tipe: http
3. Untuk masing-masing kegiatan diatas, laporkan data yang telah direkam sesuai dengan layer pada TCP/IP model. Gunakan lembar Laporan praktikum 2. TCP/IP Model. (Lampiran)

III. LAMPIRAN

3.1 Lampiran Tugas 2.1a – DNS

Laporan praktikum 2. TCP/IP Model

Kegiatan : Rekam Data Protokol DNS

Nama : Syifa Kamilla Yusron

NRP : 5325600028

Data layer

Application layer

```
Domain Name System (query)
    Transaction ID: 0x0006
    Flags: 0x0100 Standard query
        0... .... .... .... = Response: Message is a query
        .000 0... .... .... = Opcode: Standard query (0)
        .... ..0. .... .... = Truncated: Message is not truncated
        .... ...1 .... .... = Recursion desired: Do query recursively
        .... .... .0.. .... = Z: reserved (0)
        .... .... ...0 .... = Non-authenticated data: Unacceptable
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
        www.pens.ac.id: type A, class IN
            Name: www.pens.ac.id
            [Name Length: 14]
            [Label Count: 4]
            Type: A (1) (Host Address)
            Class: IN (0x0001)
    [Response In: 327652]
```

Host-to-host layer

```
User Datagram Protocol, Src Port: 58194, Dst Port: 53
    Source Port: 58194
    Destination Port: 53
    Length: 40
    Checksum: 0xde6b [unverified]
    [Checksum Status: Unverified]
    [Stream index: 7813]
    [Stream Packet Number: 1]
    [Timestamps]
        [Time since first frame: 0.000000000 seconds]
        [Time since previous frame: 0.000000000 seconds]
    UDP payload (32 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.135.10, Dst: 10.252.128.3
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 60
    Identification: 0xcdd5 (52693)
  000. .... = Flags: 0x0
      0... .... = Reserved bit: Not set
      .0.. .... = Don't fragment: Not set
      ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x4fd6 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.252.135.10
    Destination Address: 10.252.128.3
    [Stream index: 191]
```

Network Interface layer

```
Ethernet II, Src: FNLINKTECHNO_42:4f:bc (78:be:81:42:4f:bc), Dst: VMware_9f:e1:85 (00:50:56:9f
  Destination: VMware_9f:e1:85 (00:50:56:9f:e1:85)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: FNLINKTECHNO_42:4f:bc (78:be:81:42:4f:bc)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 268]
```

3.2 Lampiran Tugas 2.1b (FTP)

Laporan praktikum 2. TCP/IP Model

Kegiatan : Rekam Data Protokol FTP

Nama : Syifa Kamilla Yusron

NRP : 5325600028

Data layer

Application layer

```
File Transfer Protocol (FTP)
  PASS \r\n
      Request command: PASS
  [Current working directory: ]
```

Host-to-host layer

```
Transmission Control Protocol, Src Port: 65254, Dst Port: 21, Seq: 31, Ack: 166, Len: 7
    Source Port: 65254
    Destination Port: 21
    [Stream index: 183]
    [Stream Packet Number: 12]
  [Conversation completeness: Incomplete, DATA (15)]
      ..0. .... = RST: Absent
      ...0 .... = FIN: Absent
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ··DASS]
    [TCP Segment Len: 7]
    Sequence Number: 31    (relative sequence number)
    Sequence Number (raw): 1728356428
    [Next Sequence Number: 38    (relative sequence number)]
    Acknowledgment Number: 166    (relative ack number)
    Acknowledgment number (raw): 2305668480
    0101 .... = Header Length: 20 bytes (5)
  Flags: 0x018 (PSH, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 1... = Push: Set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····AP···]
    Window: 8027
    [Calculated window size: 8027]
    [Window size scaling factor: 1]
    Checksum: 0xcc2d [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  [Timestamps]
```

```
▾ [Timestamps]
      [Time since first frame in this TCP stream: 31.723267600 seconds]
      [Time since previous frame in this TCP stream: 4.831420800 seconds]
▾ [SEQ/ACK analysis]
      [iRTT: 4.247400 milliseconds]
      [Bytes in flight: 7]
      [Bytes sent since last PSH flag: 7]
   [Client Contiguous Streams: 1]
   [Server Contiguous Streams: 1]
   TCP payload (7 bytes)
```

Internetwork layer

```
▾ Internet Protocol Version 4, Src: 10.252.135.10, Dst: 202.9.85.80
   0100 .... = Version: 4
   .... 0101 = Header Length: 20 bytes (5)
▾ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
   Total Length: 47
   Identification: 0x0fd3 (4051)
▾ 010. .... = Flags: 0x2, Don't fragment
      0... .... = Reserved bit: Not set
      .1.. .... = Don't fragment: Set
      ..0. .... = More fragments: Not set
   ...0 0000 0000 0000 = Fragment Offset: 0
   Time to Live: 128
   Protocol: TCP (6)
   Header Checksum: 0x3996 [validation disabled]
   [Header checksum status: Unverified]
   Source Address: 10.252.135.10
   Destination Address: 202.9.85.80
   [Stream index: 1223]
```

Network Interface layer

```
▾ Ethernet II, Src: FNLINKTECHNO_42:4f:bc (78:be:81:42:4f:bc), Dst: VMware_b5:55:03 (00:50:56:b5:55:03)
  ▾ Destination: VMware_b5:55:03 (00:50:56:b5:55:03)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  ▾ Source: FNLINKTECHNO_42:4f:bc (78:be:81:42:4f:bc)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 1094]
```

3.3 Lampiran Tugas 2.1c (HTTP)

Laporan praktikum 2. TCP/IP Model

Kegiatan : Rekam Data Protokol HTTP

Nama : Syifa Kamilla Yusron

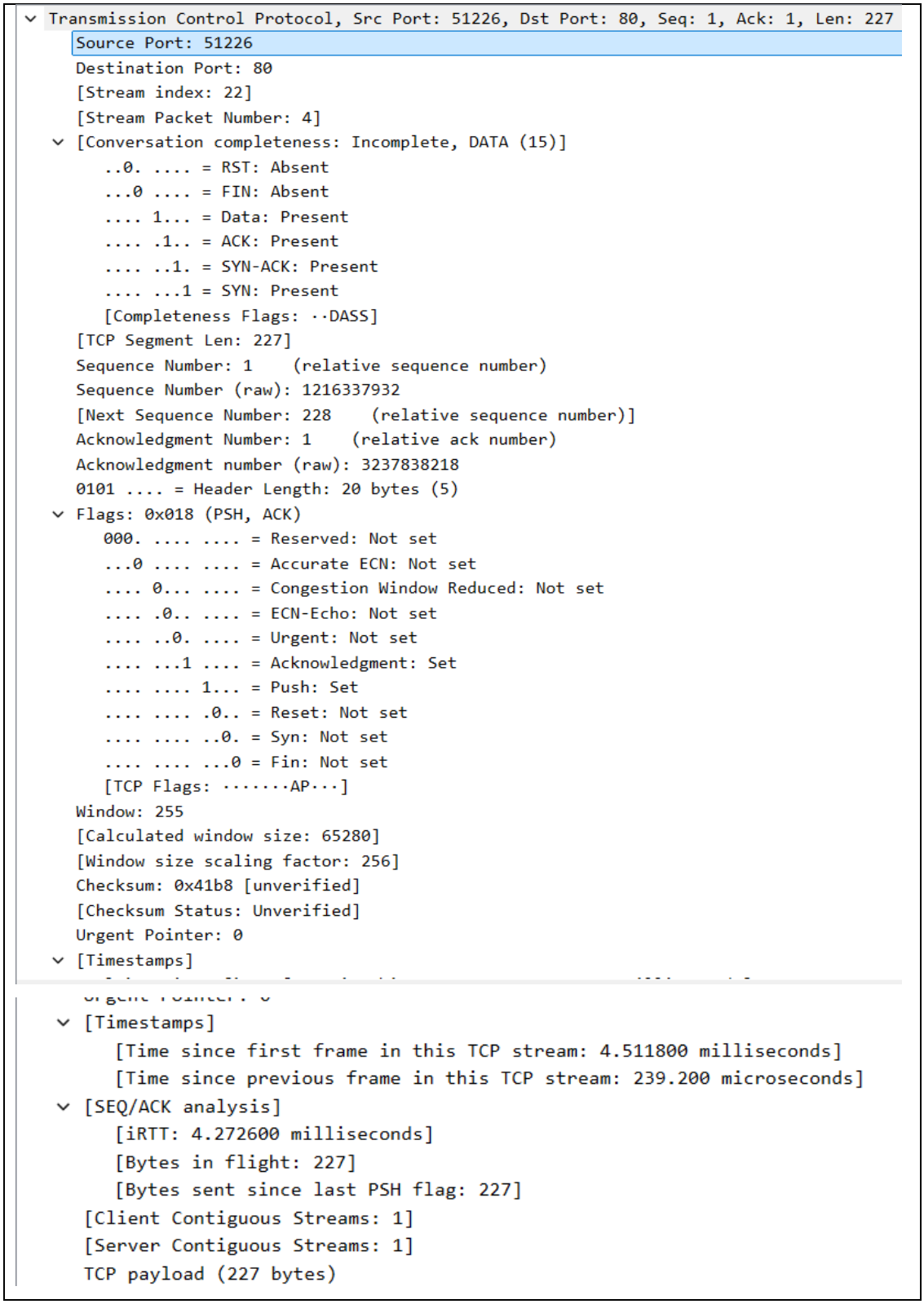
NRP : 5325600028

Data layer

Application layer

```
Hypertext Transfer Protocol
  GET / HTTP/1.1\r\n
      Request Method: GET
      Request URI: /
      Request Version: HTTP/1.1
    Cache-Control: max-age = 3600\r\n
    Connection: Keep-Alive\r\n
    Accept: */*\r\n
    If-Modified-Since: Wed, 13 May 2026 23:14:58 GMT\r\n
    If-None-Match: "6a0505f2-328"\r\n
    User-Agent: Microsoft-CryptoAPI/10.0\r\n
    Host: x1.c.lencr.org\r\n
    \r\n
    [Response in frame: 4512]
    [Full request URI: http://x1.c.lencr.org/]
```

Host-to-host layer



Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.135.10, Dst: 23.32.245.29
   0100 .... = Version: 4
   .... 0101 = Header Length: 20 bytes (5)
 ⌄ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
   Total Length: 267
   Identification: 0xb507 (46343)
 ⌄ 010. .... = Flags: 0x2, Don't fragment
      0... .... = Reserved bit: Not set
      .1.. .... = Don't fragment: Set
      ..0. .... = More fragments: Not set
   ...0 0000 0000 0000 = Fragment Offset: 0
   Time to Live: 128
   Protocol: TCP (6)
   Header Checksum: 0xa6a1 [validation disabled]
   [Header checksum status: Unverified]
   Source Address: 10.252.135.10
   Destination Address: 23.32.245.29
   [Stream index: 253]
Transmission Control Protocol, Src Port: 51226, Dst Port: 80, Seq: 1, Ack: 1, Len: 2
```

Network Interface layer

```
⌄ Ethernet II, Src: FNLINKTECHNO_42:4f:bc (78:be:81:42:4f:bc), Dst: VMware_9f:e1:85 (00:50:56:9f:e1:85)
  ⌄ Destination: VMware_9f:e1:85 (00:50:56:9f:e1:85)
       .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
       .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  ⌄ Source: FNLINKTECHNO_42:4f:bc (78:be:81:42:4f:bc)
       .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
       .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 55]
```